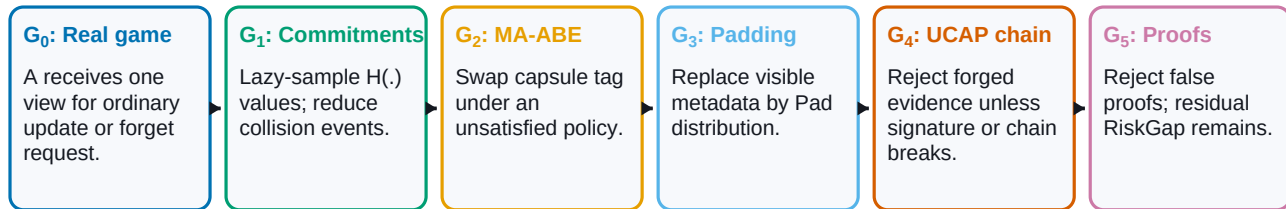


Policy-authenticated update-hiding game and reduction purpose

Each hop bounds one observable advantage: commitments, policy capsule, channel shape, audit forgery, and false proofs.



Reduction bound

$$\text{Adv}_{\text{auth}}(A) \leq \text{Adv}_{\text{MA-ABE}}(B_1) + \text{Adv}_{\text{RO}}(B_2) + \text{Adv}_{\text{PAD}}(B_3) + \text{Adv}_{\text{SIG}}(B_4) \\ + \text{Adv}_{\text{ZK}}(B_5) + \epsilon_R + \text{negl}(\lambda)$$

B_1 : policy capsule indistinguishability

B_2 : commitment consistency

B_3 : request-type padding

B_4 : audit-chain unforgeability

B_5 : authorization/repair proof soundness